

MAYUKH DAS

linkedin.com/in/mayukh-das-a38319148
Canberra, ACT, Australia | github.com/Mayukh-D

EDUCATION

Australian National University, Master of Computing

Feb 2025 – Present

Relevant coursework: Deep Learning (CNNs, RNNs, Transformers, Generative Models), Advanced ML / Generative AI (Diffusion Models, LLMs), Statistical Machine Learning, Human-Computer Interaction, Software Engineering, Software Construction (Design Patterns, Android), Structured Programming (Java), Logic & Computation.

Manipal Institute of Technology, B.Tech, Automobile Engineering

Aug 2017 – Sep 2021

SKILLS

Languages: Python, Java, C#, JavaScript, SQL

ML / AI: Deep Learning, Transformers, Diffusion Models, LLMs, Explainable AI (XAI), Supervised & Unsupervised Learning

Frameworks & Tools: PyTorch, scikit-learn, LLM APIs (OpenAI / GPT), Git, Jupyter, .NET, MVC, React, Node (MERN), HTML, CSS. Rated Advanced in UX Design and Machine Learning at Accenture.

WORK EXPERIENCE

Accenture

Jan 2022 – Feb 2025

Advanced Application Engineering Analyst

- Built **Ingrain**, an AIOps tool generating predictive ML pipelines for automated IT ticket resolution, significantly reducing manual triage effort.
- Integrated Explainable AI (XAI) and dynamic model performance reporting to improve client interpretability of predictions; designed and implemented the supporting user interfaces from scratch.
- Built **Quasar++**, a ChatGPT API-based file-querying chatbot, enabling natural-language access to internal documents.
- Rated Advanced in UX Design and Machine Learning under Accenture's internal competency framework.

Cognizant

Oct 2021 – Jan 2022

Programmer Analyst

- Designed and implemented full-stack web systems using .NET, MVC, React, HTML, CSS, and JavaScript, building all UI from scratch; delivered a Hospital Logistics System to manage COVID-19 operational workflows.

PROJECTS

GPT Implementation (nanoGPT) — *PyTorch · CUDA · Transformer Architecture*

- Implemented a GPT language model from scratch in PyTorch, covering attention mechanisms, positional encoding, and autoregressive text generation.

Diffusion Model Implementation — *PyTorch · CUDA · Generative AI*

- Built a diffusion model from scratch covering score-based generative modelling, noise scheduling, and denoising mathematics, as part of ANU Advanced ML coursework.

FoodLens — Smart Food Scanner PWA — *HTML · CSS · JavaScript · Web APIs · PWA*

- Built a Progressive Web App for sustainable food shopping: real-time barcode scanning via BarcodeDetector API, live product lookup via Open Food Facts, allergen/expiry/carbon flagging, and A/B interaction logging for HCI research (UN SDG 12 & 3).

Yuma — Android Social Platform — *Java · Android · DAO · Singleton · Factory · Iterator*

- Co-built a native Android social media app (post feed, reactions, DMs, admin moderation) applying Singleton, Factory, and Iterator design patterns across a layered DAO architecture with full CSV persistence.

STUDENT ROLES

ANU College of Systems and Society

2025 – Present

Student Ambassador

- Represent the ANU College at recruitment events and school visits; support prospective-student transition and onboarding.

ANU Students' Association (ANUSA)

2025 – Present

Postgraduate Representative, College of Systems and Society

- Elected representative for the CSS postgraduate cohort; advocate for student interests and liaise with faculty and administration.

RESEARCH PUBLICATIONS

- Das, M. (2024). *An Overview on the Role of AI in Modern Advancements of Material Science*. ES General, Vol. 5. DOI 10.30919/esg1183. First author.
- Karthik, A., Das, M. et al. (2021). *Electric Propulsion for Fixed Wing Aircrafts: Classifications, Designs, and Challenges*. Engineered Science, Vol. 16, pp. 129–145. DOI 10.30919/es8d573. Scopus Q1.
- Momaya, V.J., Das, M. et al. (2023). *Revolutionizing Organ Donation With Blockchain Technology*. J. of Computers, Mechanical and Management, Vol. 2(5), pp. 31–38. DOI 10.57159/gadl.jcmm.2.5.23071. Scopus-indexed.
- Conference presentation:** Work-in-progress form of the Material Science paper, National Conference on Recent Trends in Computers, Mechanical and Management (RTCMM), 2023.